

Quiz (Habanero)

- Q1.** A rain barrel contains 15 gallons of water with 0.05 concentration of salt. How many gallons of salt-free water should be added to achieve 0.01 concentration?
(A) 5 gallons
(B) 10 gallons
(C) 15 gallons
(D) 20 gallons
(E) 25 gallons
- Q2.** A pollution monitoring station measures 30
(A) 9
(B) 12
(C) 18
(D) 21
(E) 28
- Q3.** A weather station uses a mixture of $\frac{2}{3}$ water and $\frac{1}{3}$ antifreeze to prevent pipes from freezing. If the station uses 24 gallons of the mixture, how many gallons of antifreeze are used?
(A) 6 gallons
(B) 8 gallons
(C) 10 gallons
(D) 12 gallons
(E) 16 gallons
- Q4.** A forest ecosystem requires a ratio of 3 : 5 of nitrogen to phosphorus for healthy plant growth. If a fertilizer contains 12
(A) 10
(B) 15
(C) 20
(D) 25
- (E) 30
- Q5.** A city's water supply is 80
(A) 0.022
(B) 0.025
(C) 0.028
(D) 0.032
(E) 0.035
- Q6.** A greenhouse maintains a temperature of $75^{\circ}F$ during the day and $50^{\circ}F$ at night. If the outside temperature is $40^{\circ}F$, what percentage of the greenhouse's heating system is used at night?
(A) 25
(B) 33
(C) 40
(D) 50
(E) 60
- Q7.** A region experiences 60
(A) 68
(B) 72
(C) 76
(D) 80
(E) 84
- Q8.** A meteorologist measures 10 mm of rainfall in a region with 20
(A) 25
(B) 30
(C) 35
(D) 40
(E) 45

Open Ended Questions (FRQ)

- Q9.** A city's water supply is a mixture of 40
- Q10.** A weather station monitors the temperature in a region, which follows a ratio of 2 : 3 of daytime temperature to nighttime temperature. If the daytime temperature is $85^{\circ}F$ and the nighttime temperature is $x^{\circ}F$, find the value of x and explain the mathematical process used to solve the problem.

Chef's Tips

- Check calculations carefully
- Watch for unit conversions
- Use algebraic methods when possible